# Experiment No. 10

**Aim:** To study the operation of DC - AC converter (Inverter) using R-L load. To observe the output waveforms (Simulation) using MATLAB Simulink.

**Simulation tools required:**

MATLAB 2015a

**Learning outcomes:**

**Theory :**

A power inverter, or inverter, is an electronic device or circuitry that changes direct current (DC) to alternating current (AC). The input voltage, output voltage and frequency, and overall power handling depend on the design of the specific device or circuitry. The inverter does not produce any power; the power is provided by the DC source.

The input to the inverter is a direct dc source or dc source derived from an AC source. For example, the primary source of input power may be utility AC voltage supply that is converted to DC by a rectifier with a filter capacitor and then 'inverted' back to AC using an inverter. Here, the final AC output may be of a different frequency and magnitude than the input AC of the utility supply. A voltage source is called *stiff*, if the source voltage magnitude does not depend on the load connected to it. All voltage source inverters assume a stiff voltage supply at the input.

If the input is a voltage source, then the inverter is called a Voltage Source Inverter (VSI). Similarly, in a Current Source Inverter (CSI), the input to the circuit is a current source. The VSI circuit has direct control over 'the output (AC) voltage' whereas a CSI directly controls 'the output (AC) current'.

Figure.1 shows the power circuit diagram for single-phase bridge voltage source inverter. In this four switches (in 2 legs) are used to generate the AC waveform at the output. Semiconductor switches like IGBT, MOSFET or BJT are used. Four switches are sufficient for a resistive load as the load current $i_o$ is in phase with the output voltage vo. However, in the case of RL loads where $i_o$ is not in phase with $v_o$, diodes are connected in anti-parallel with the switches which allow the conduction of the current in the reverse direction even when the switch is turned OFF. These diodes are called *Feedback Diodes* since the energy is fed back to the (DC) source.

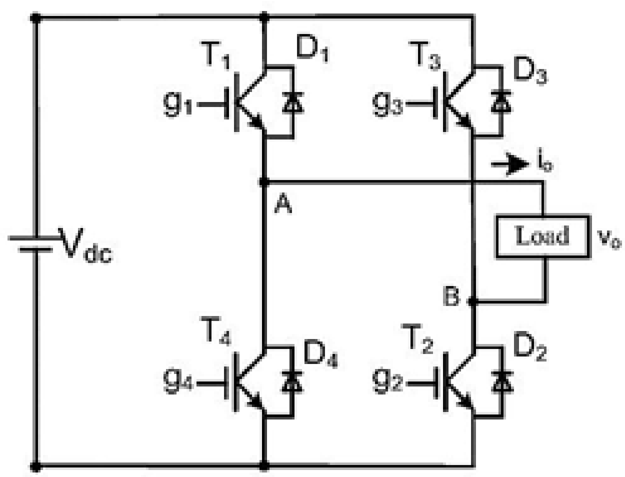
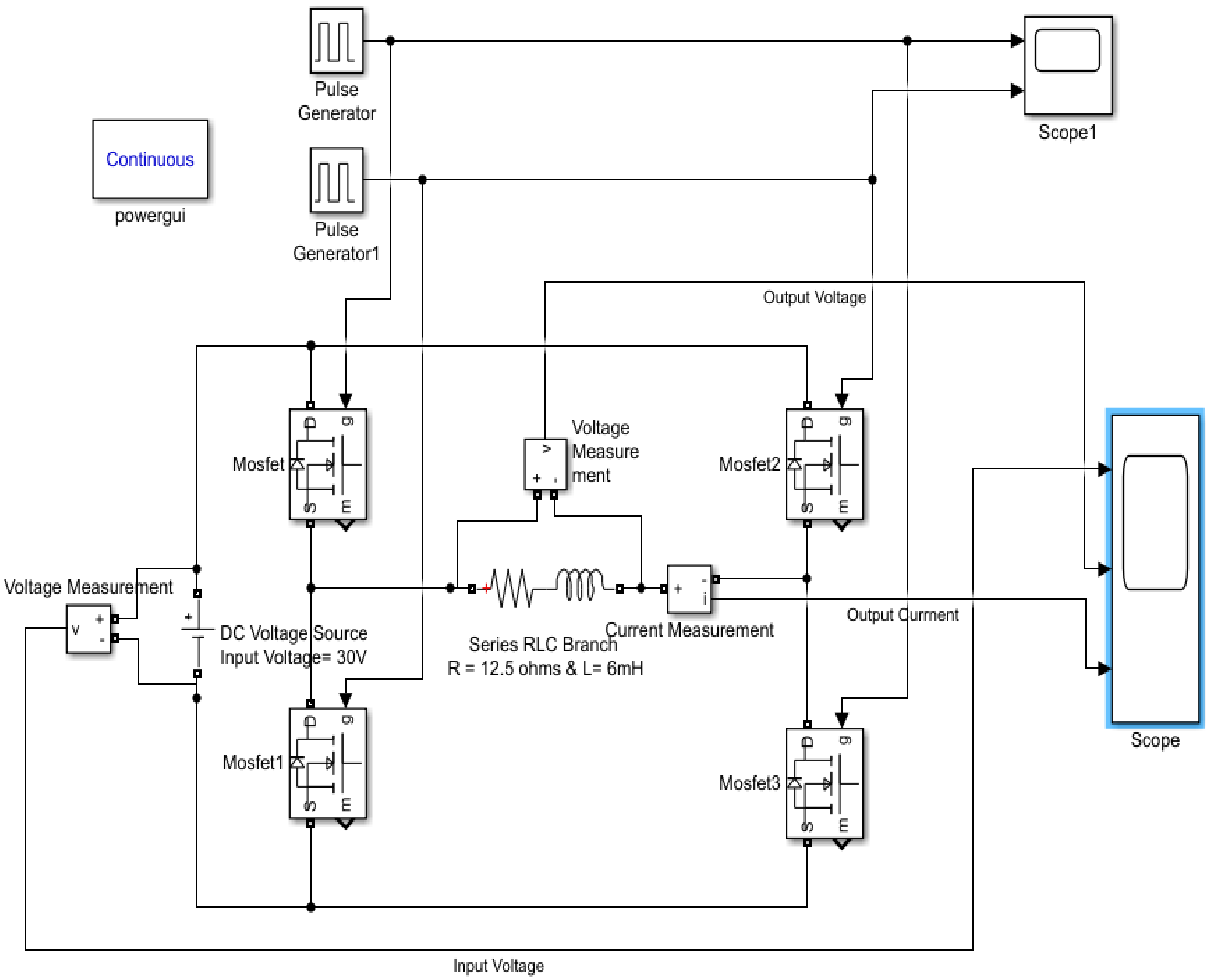



**Figure.1**

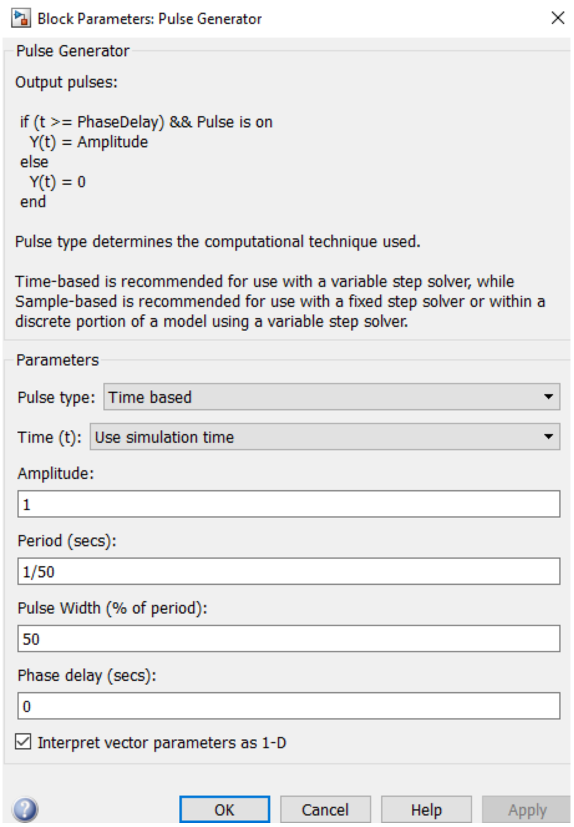
A typical power inverter requires a relatively stable DC power source capable of supplying enough current for the intended power demands of the system. The input voltage depends on the design and purpose of the inverter.

I. **Square Wave Modulation:** for **R = 12.5 ohms**, **L = 6mH**, **DC voltage source of 30V**

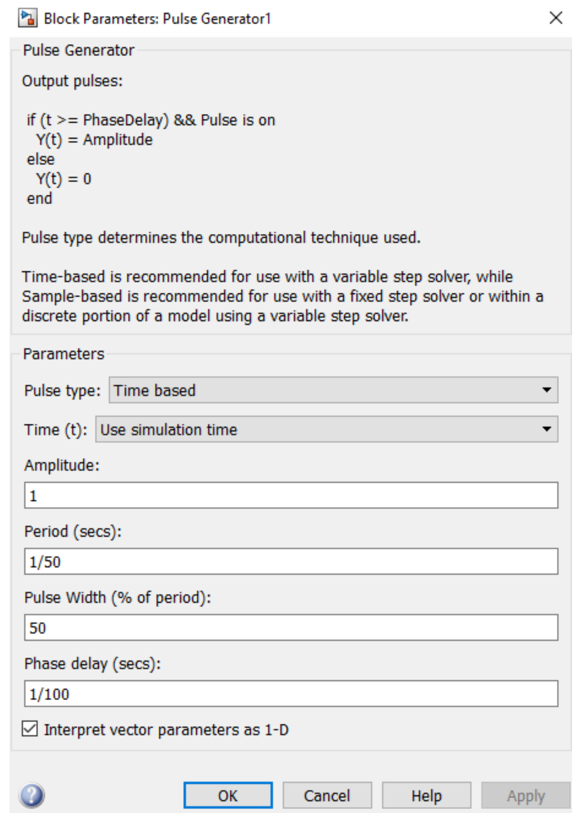


## Pulse generators showing square wave pulses

1. **Pulse generator**



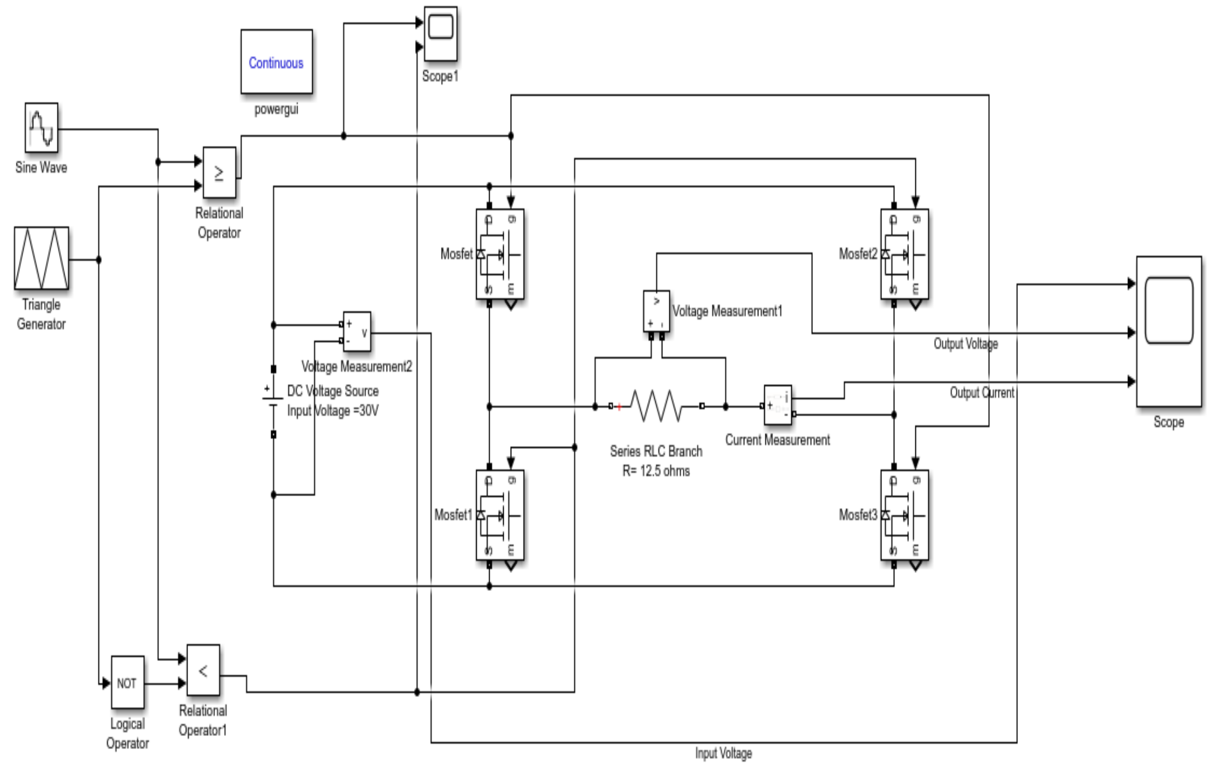

2. **Pulse Generator 1**




### Square Wave Observation table:

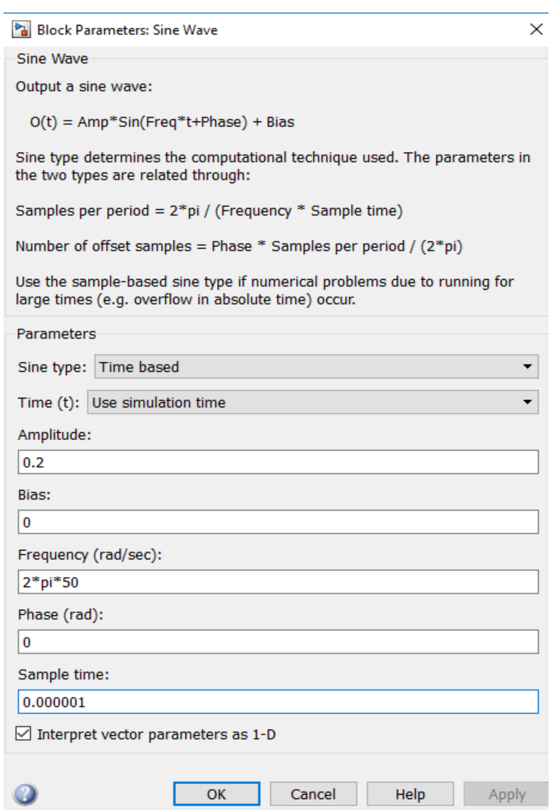
| S. No. | Input Voltage | Output voltage Fundamental (RMS) | Output voltage THD | Output Voltage (RMS) At 150Hz | Output Current Fundamental (RMS) | Output Current (RMS) At 150 Hz | Output Current THD |
|---|---|---|---|---|---|---|---|
| (Square Wave ) | 30V | | | | | | |

## II. <u>Sinewave Generation:</u> for **R = 12.5 ohms, , DC voltage source of 30V**




## <u>Block parameters for generating sine PWM:</u>

### 1. Sine Wave for 0.2 Modulation Index (M.I.)



Block Parameters: Sine Wave

Sine Wave

Output a sine wave:

O(t) = Amp*Sin(Freq*t+Phase) + Bias

Sine type determines the computational technique used. The parameters in the two types are related through:

Samples per period = 2*pi / (Frequency * Sample time)

Number of offset samples = Phase * Samples per period / (2*pi)

Use the sample-based sine type if numerical problems due to running for large times (e.g. overflow in absolute time) occur.

Parameters

Sine type: Time based

Time (t): Use simulation time

Amplitude:

0.2

Bias:

0

Frequency (rad/sec):

2*pi*50

Phase (rad):

0

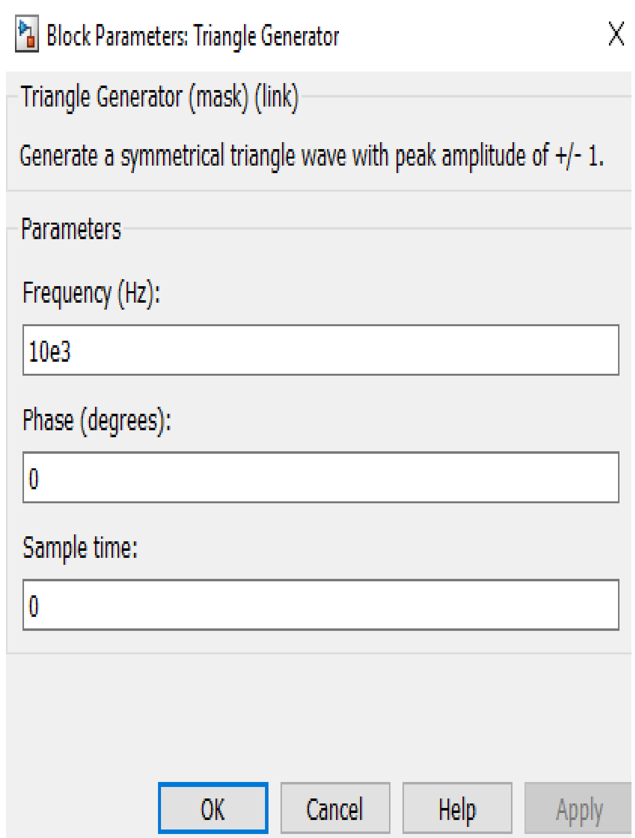
Sample time:

0.000001

☑ Interpret vector parameters as 1-D

OK Cancel Help Apply

### 2. Triangle Generator



Block Parameters: Triangle Generator

Triangle Generator (mask) (link)

Generate a symmetrical triangle wave with peak amplitude of +/- 1.

Parameters

Frequency (Hz):

10e3

Phase (degrees):

0

Sample time:

0

OK Cancel Help Apply

**Sinewave Observation table:**

| S. No. | Input Voltage | Output voltage Fundamental (RMS) | Output voltage THD | Output Voltage (RMS) At 150Hz | Output Current Fundamental (RMS) | Output Current (RMS) At 150 Hz | Output Current THD |
|---|---|---|---|---|---|---|---|
| (SineWave ) MI = 0.2 | 30V | | | | | | |
| (Sine Wave) MI = 0.6 | 30V | | | | | | |
| (Sine Wave) MI = 1 | 30V | | | | | | |

**Both the Simulation and Hardware results have to be attached in the PE Lab Results folder “Exp10_Part B.doc” file.**